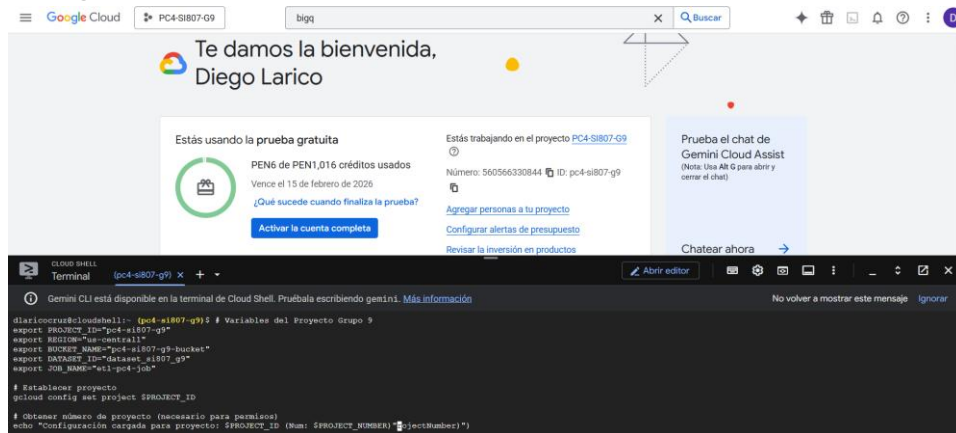


DESARROLLO COMPLETO DE LA PC4

Nombres: Diego Larico Cruz (20212679A)

Gabriel Cabana Cazani (20212541J)

1. Configurar nombres de variables



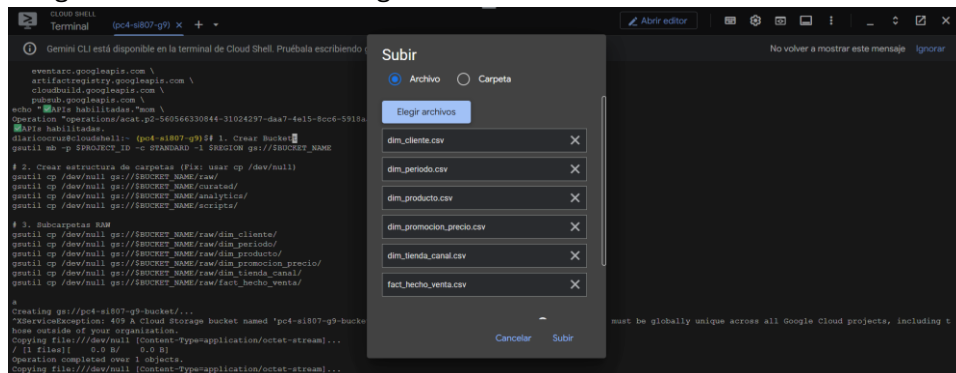
2. Habilitar APIS



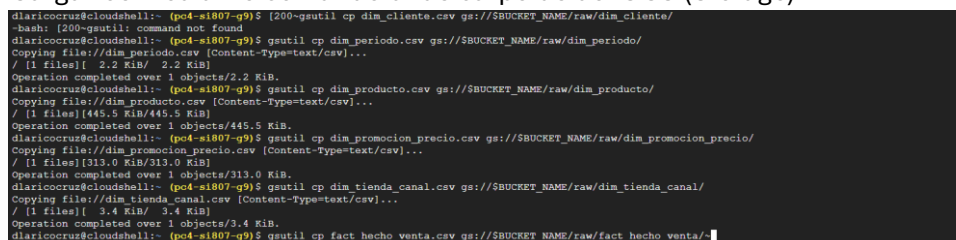
3. Crear directorio de carpetas



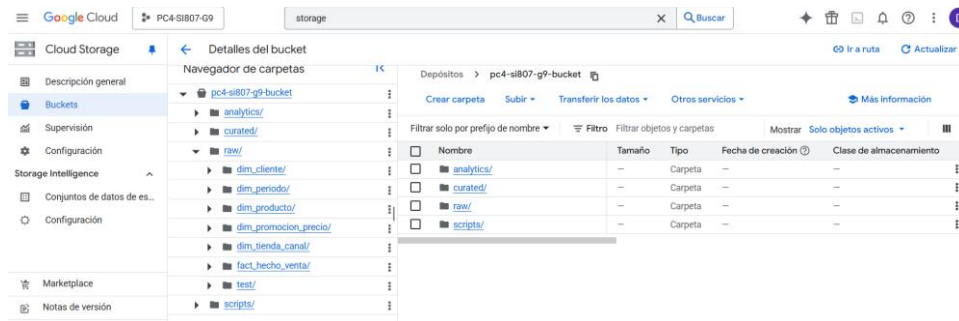
4. Cargar los archivos en el Google Shell



5. Cargarlos mediante comando a las carpetas del GCS (Storage)

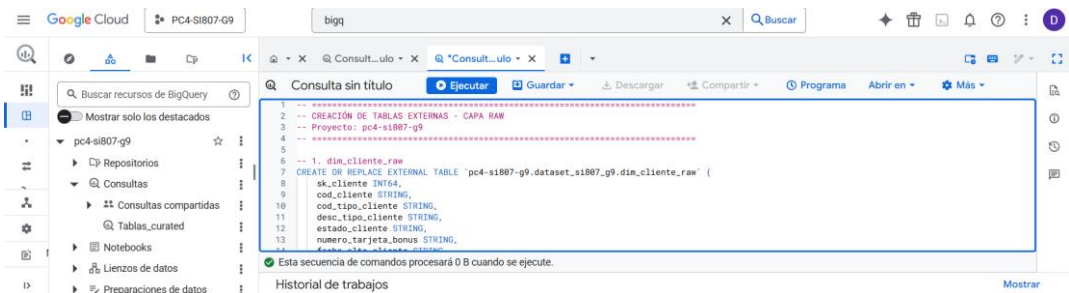


6. Verificamos

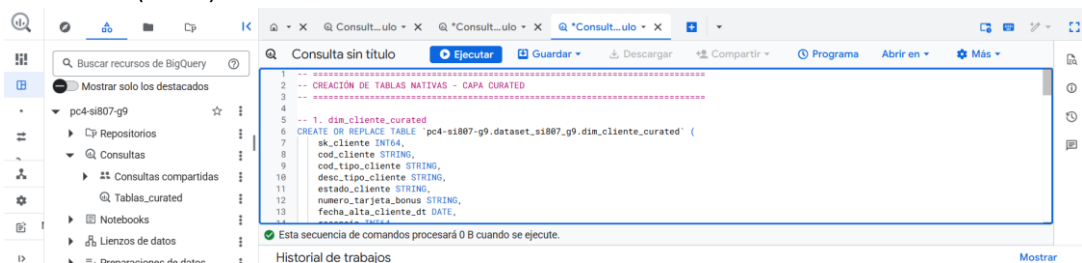


7. En BIG QUERY, ejecutamos las sentencias SQL para creación de tablas RAW CURATED y RESUMEN

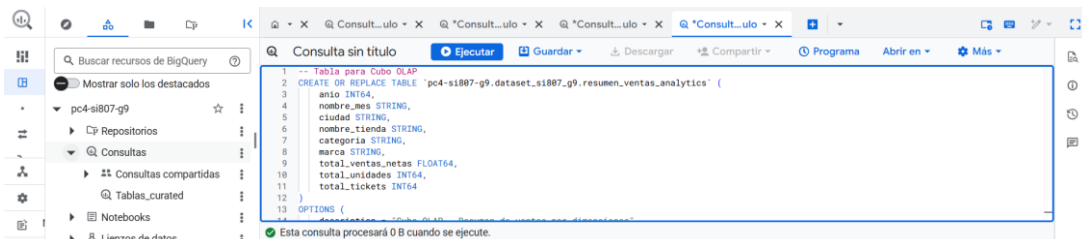
- RAW



- CURATED (vacía)



- ANALYTICS



8. Creamos el archivo etl-curated-job.py

```
dlaricocruz@cloudshell:~ (pc4-s1807-g9) $ nano etl-curated-job2.py
dlaricocruz@cloudshell:~ (pc4-s1807-g9) $ gsutil cp etl-curated-job2.py gs://$BUCKET_NAME/scripts/
Copying file://etl-curated-job2.py [Content-Type=text/x-python]...
/ [1 files][ 11.2 KiB/ 11.2 KiB]
```

9. Ejecutamos el Job Dataproc

```
Cloud Shell
Terminal (pc4-si807-g9) x +
Gemini CLI está disponible en la terminal de Cloud Shell. Pruébala escribiendo gemini. Más información
No volver a mostrar este mensaje Ignorar

gcloud info --run-diagnostics
diagnostic@cloudshell:~$ (pc4-si807-g9)
gcloud dataproc batches submit pyspark \
  --project=SI807-G9 \
  --region=EUROPE \
  --batch=${JOB_NAME}-v2-2 (data +tkbhd-tkbhd) \
  --version=1.1 \
  --submitter=spark \
  --jar=gs://spark-lib/bigquery/spark-bigquery-with-dependencies_2.12-0.32.2.jar \
  --properties=spark.executor.cores=4,spark.executor.memory=16g,spark.executor.instances=2,spark.dynamicallocation.enabled=false \
  gs://SI807-G9/etl/etl-jobs/etl-jobs-2.py
Batch (etl-pc4-job-v2-20251201-02019) submitted.
Using the default container image
Waiting for container log creation
PYSHELL_PYTHON=/opt/dataproc/conda/bin/python
JAVA_HOME=/usr/lib/jvm/temurin-11-jdk-amd64
SPARK_EXTRA_CLASSPATH
Loading settings: file = /etc/spark/conf/ivysettings.xml
*****
INICIANDO ETL-PC4-JOB - GRUPO 9
*****
25/12/01 02:04:51 ERROR SparkContext: Failed to add gs://spark-lib/bigquery/spark-bigquery-with-dependencies_2.12.jar to Spark environment
java.io.FileNotFoundException: Jar gs://spark-lib/bigquery/spark-bigquery-with-dependencies_2.12.jar not found
at org.apache.spark.SparkContext.addJar(SparkContext.scala:2034)
at org.apache.spark.SparkContext.addJar(SparkContext.scala:2077)
at org.apache.spark.SparkContext.$anonfun$show14(SparkContext.scala:1450)
at org.apache.spark.SparkContext.$anonfun$show14$adapted(SparkContext.scala:1450)
at scala.collection.mutable.ResizableArray.foreach(ResizableArray.scala:62)
at scala.collection.mutable.ResizableArray.foreach(ResizableArray.scala:55)
at scala.collection.mutable.ArrayBuffer.foreach(ArrayBuffer.scala:49)
at org.apache.spark.SparkContext.$anonfun$show14(SparkContext.scala:1450)
at org.apache.spark.SparkContext.$anonfun$show14$adapted(SparkContext.scala:1450)
at java.base/jdk.internal.reflect.NativeConstructorAccessorImpl.newInstance0(Native Method)
at java.base/jdk.internal.reflect.NativeConstructorAccessorImpl.newInstance(NativeConstructorAccessorImpl.java:42)
at java.base/jdk.internal.reflect.DelegatingConstructorAccessorImpl.newInstance(DelegatingConstructorAccessorImpl.java:45)
```

10. Revisamos

```
Cloud Shell
Terminal (pc4-si807-g9) x +
Gemini CLI está disponible en la terminal de Cloud Shell. Pruébala escribiendo gemini. Más información

at java.base/java.lang.Thread.run(Thread.java:829)
[INFO] Sesión Spark iniciada: 3.3.4
[INFO] Cargando dim_cliente desde gs://pc4-si807-g9-bucket/raw/dim_cliente/*.csv
[SUCCESS] dim_cliente cargada: 12096 registros
[INFO] Transformando dim_cliente...
[SUCCESS] dim_cliente transformada: 12096 registros
[INFO] Escribiendo dim_cliente a BigQuery: pc4-si807-g9.dataset_si807_g9.dim_cliente_curated
[SUCCESS] dim_cliente escrita exitosamente
[INFO] Cargando dim_periodo desde gs://pc4-si807-g9-bucket/raw/dim_periodo/*.csv
[SUCCESS] dim_periodo cargada: 48 registros
[INFO] Transformando dim_periodo...
[SUCCESS] dim_periodo transformada: 48 registros
[INFO] Escribiendo dim_periodo a BigQuery: pc4-si807-g9.dataset_si807_g9.dim_periodo_curated
[SUCCESS] dim_periodo escrita exitosamente
[INFO] Cargando dim_producto desde gs://pc4-si807-g9-bucket/raw/dim_producto/*.csv
[SUCCESS] dim_producto cargada: 6048 registros
[INFO] Transformando dim_producto...
[SUCCESS] dim_producto transformada: 6048 registros
[INFO] Escribiendo dim_producto a BigQuery: pc4-si807-g9.dataset_si807_g9.dim_producto_curated
[SUCCESS] dim_producto escrita exitosamente
[INFO] Cargando dim_promocion desde gs://pc4-si807-g9-bucket/raw/dim_promocion_precio/*.csv
[SUCCESS] dim_promocion cargada: 6048 registros
[INFO] Transformando dim_promocion_precio...
[SUCCESS] dim_promocion transformada: 6048 registros
[INFO] Escribiendo dim_promocion a BigQuery: pc4-si807-g9.dataset_si807_g9.dim_promocion_precio_curated
[SUCCESS] dim_promocion escrita exitosamente
[INFO] Cargando dim_tienda desde gs://pc4-si807-g9-bucket/raw/dim_tienda_canal/*.csv
[SUCCESS] dim_tienda cargada: 50 registros
[INFO] Transformando dim_tienda_canal...
[SUCCESS] dim_tienda transformada: 50 registros
[INFO] Escribiendo dim_tienda a BigQuery: pc4-si807-g9.dataset_si807_g9.dim_tienda_canal_curated
[SUCCESS] dim_tienda escrita exitosamente
[INFO] Cargando fact_venta desde gs://pc4-si807-g9-bucket/raw/fact_hecho_venta/*.csv
[SUCCESS] fact_venta cargada: 100000 registros
[INFO] Transformando fact_hecho_venta...
```

11. Con ello ya podemos hacer la conexión con Looker Studio

Añadir datos al informe

Credenciales de acceso a datos: Diego Larico

PROYECTOS RECIENTES

MIS PROYECTOS

PROYECTOS COMPARTIDOS

CONSULTA PERSONALIZADA

CONJUNTOS DE DATOS PÚBLICOS

Project

Search Projects

Enter Project Id manually

PC4-SI807-G9

ETL-Retail-Analytics

SI807-G9

My First Project

copla1 kampo

Proyecto-Kampo-V1

Demo-Kampo

Conjunto de datos

Search Datasets

dataset_si807_g9

Table

Search Tables

dim_periodo_curated

dim_periodo_raw

dim_producto_curated

dim_producto_raw

dim_promocion_precio_curated

dim_promocion_precio_raw

dim_tienda_canal_curated

dim_tienda_canal_raw

fact_hecho_venta_curated

fact_hecho_venta_raw

resumen_ventas_analytics

Cancelar

Añadir

Dashboard

Restablecer

Compartir

Editar

	desc_canal	cadena	Ticket Promedio
1.	PRESENCIAL	METRO	S/ 106,85
2.	PRESENCIAL	WONG	S/ 107,29
3.	ONLINE	METRO	S/ 108,81
4.	ONLINE	WONG	S/ 107,2

1 - 4 / 4 < >

Margen Promedio por Ticket

	cadena	desc_c...	%Descuento sobre Venta ...
1.	WONG	PRESEN...	
2.	WONG	ONLINE	
3.	METRO	ONLINE	
4.	METRO	PRESEN...	

1 - 5 / 5 < >

% de Tickets con promoción

Con eso terminamos la parte de funciones básicas

FUNCIONES AVANZADAS

CLOUD FUNCTIONS + PUB SUB

Lo que buscamos es realizar es automatizar la ejecución del Job cuando se carga un nuevo archivo en las carpetas de raw

```
dlaricocruz@cloudshell:~ (pc4-si807-g9)$ gcloud pubsub topics create etl-notifications
```

Mis paneles + Etiquetas Borrar Etiquetas Administrar etiquetas

Filtro Buscar entre tus paneles

Todos 9	☐	☐	BigQuery Continuous Query Monitoring	Servicios de Google	This dashboard has charts displaying BigQuery metrics for a continuous query.	
Vistos recientemente 1	☐	☐	Cloud Run Monitoring	Servicios de Google	This dashboard has 7 charts for the related Cloud Run metrics, including Request...	
Favoritos 0	☐	☐	Cloud Storage	Servicios de Google	View GCS Bucket metrics such as Requests, Network Traffic, Object Count and...	
Recomendado 8	☐	☐	Firewalls	Servicios de Google	This dashboard shows the list of firewalls in a project.	
Integraciones	☐	☐	Google Cloud Storage	En contexto	View GCS Bucket metrics such as Requests, Network Traffic, Object Count and...	
Agente de operacio... 0	☐	☐	Logs Dashboard	Servicios de Google	This dashboard has charts displaying log-based metrics for GKE Clusters, GKE...	
GMP 0	☐	☐	New Dashboard - nov 30, 2025 10:46 p.m.	Personalizado		
	☐	☐	Pub/Sub	Servicios de Google	View topic metrics for published message count and average message size, as...	

Configuramos los roles IAM para poder correr la función

```
dlaricocruz@cloudshell:~ (pc4-si807-g9)$ # Permiso al Agente de Storage para publicar en Pub/Sub
gcloud projects add-iam-policy-binding $PROJECT_ID \
  --member="serviceAccount:service-$PROJECT_NUMBER@gs-project-accounts.iam.gserviceaccount.com" \
  --role="roles/pubsub.publisher"

# Permiso a la Service Account de Compute para usar Dataproc
gcloud projects add-iam-policy-binding $PROJECT_ID \
  --member="serviceAccount:$PROJECT_NUMBER-compute@developer.gserviceaccount.com" \
  --role="roles/dataproc.editor"
```

Revisamos

IAM Más información

Permitir Rechazada Historial de recomendaciones

Filtro Escribir el nombre o valor de la propiedad

☐ Tipo	Principal ↑	Nombre	Rol	Estadísticas de seguridad ⓘ
☐	560566330844-compute@developer.gserviceaccount.com	Compute Engine default service account	Editor	
			Publicador de Pub/Sub	
			Visualizador de objetos de Storage	
☐	dlaricocruz@gmail.com	Diego Larico	Propietario	
☐	pc4-si807-g9@appspot.gserviceaccount.com	App Engine default service account	Editor	

Desplegamos la función

```
dlaricocruz@cloudshell:~/cloud-function-etl (pc4-si807-g9)$ # Desplegar (TARDA 3-5 MINUTOS)
gcloud functions deploy trigger-etl-on-upload \
  --gen2 \
  --runtime=python311 \
  --region=$REGION \
  --source=. \
  --entry-point=trigger_etl_on_file_upload \
  --trigger-event-filters="type=google.cloud.storage.object.v1.finalized" \
  --trigger-event-filters="bucket=$BUCKET_NAME" \
  --timeout=540s \
  --memory=512MB \
  --max-instances=5
Preparing function...done.
Updating function (may take a while)...
[Build] Logs are available at [https://console.cloud.google.com/cloud-build/builds;region=us-central1/1212b4c4-4931-43b7-88d3-c5ef26e333e6?project=560566330844]...
[Service]warning
[Trigger]done
Completed with warnings:
[INFO] A new revision will be deployed serving with 0% traffic.
You can view your function in the Cloud Console here: https://console.cloud.google.com/functions/details/us-central1/trigger-etl-on-upload?project=pc4-si807-g9

buildConfig:
  automaticUpdatePolicy: {}
  build: projects/560566330844/locations/us-central1/builds/1212b4c4-4931-43b7-88d3-c5ef26e333e6
  dockerRegistry: ARTIFACT_REGISTRY
  dockerRepository: projects/pc4-si807-g9/locations/us-central1/repositories/gcf-artifacts
  entryPoint: trigger_etl_on_file_upload
  runtime: python311
  serviceAccount: projects/pc4-si807-g9/serviceAccounts/560566330844-compute@developer.gserviceaccount.com
  source:
    storageSource:
      bucket: gcf-v2-sources-560566330844-us-central1
```

Lista Métricas

Filtro Filtrar temas

☐ ID del tema ↑	Clave de encriptación	Nombre del tema	Retención	Fuente de transferencia
☐ etl-notifications	Google-managed	projects/pc4-si807-g9/topics/...	—	—
☐ eventarc-us-central1-trigger-etl-on-upload-337918-612	Google-managed	projects/pc4-si807-g9/topics/...	—	—

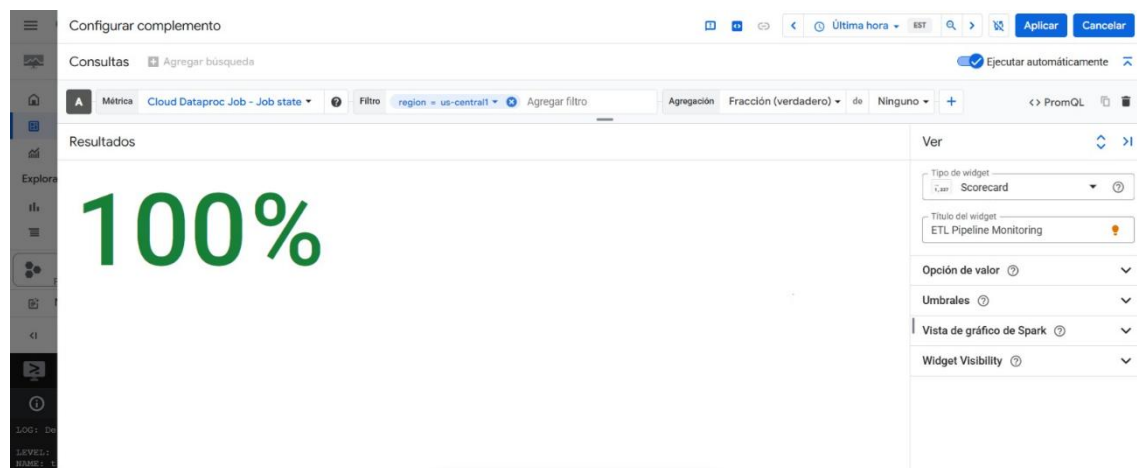
MONITORING

Lo usamos para crear alarmas

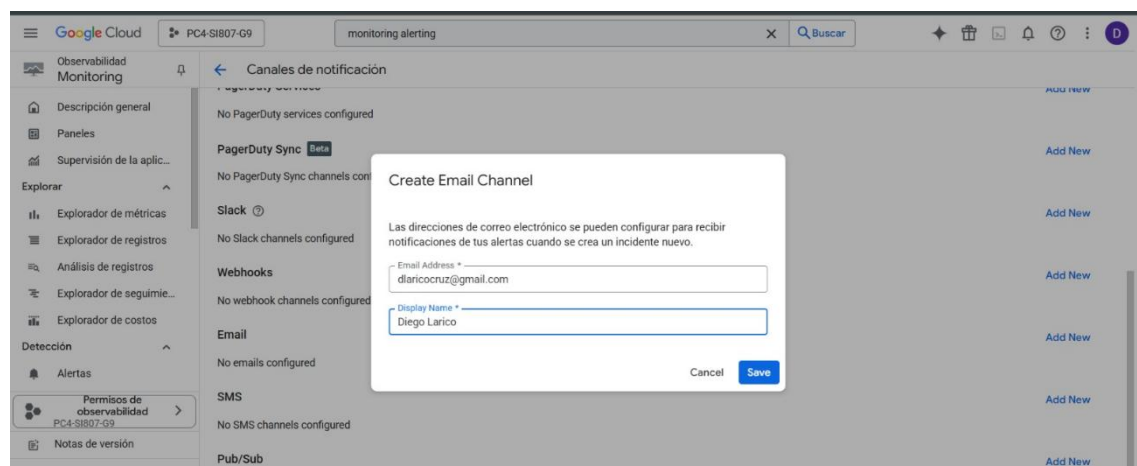
```
dlaricocruz@cloudshell:~ (pc4-si807-g9) $ cat > alert-policy.yaml << 'EOF'
displayName: "Dataproc Job Failed - Grupo 09"
combiner: OR
conditions:
  - displayName: "Job Status Failed"
    conditionThreshold:
      filter: 'resource.type="dataproc_batch" AND metric.type="dataproc.googleapis.com/batch/job/failed_count"'
      comparison: COMPARISON_GT
      thresholdValue: 0
      duration: 60s
      aggregations:
        - alignmentPeriod: 60s
          perSeriesAligner: ALIGN_MAX
alertStrategy:
  autoClose: 86400s
EOF

gcloud alpha monitoring policies create --policy-from-file=alert-policy.yaml
```

Se despliegan; por ejemplo creamos una métrica para medir la eficacia de los Jobs corridos



De manera adicional, se puede configurar para enviar notificaciones via correo



SEGURIDAD

Habilitamos KMS y creamos llaves de seguridad

```

$ in https://cloud.google.com/composer/docs/composer-2/access-control/service-account
dlaricacruz@cloudshell:~ (pc4-si807-g9) $ # 1. Habilitar API de KMS
gcloud services enable cloudkms.googleapis.com

# 2. Crear un "llavero" (KeyRing) en la región
gcloud kms keyrings create keyring-retail-g9 \
  --location=$REGION

# 3. Crear la Llave de Encriptación (Key)
gcloud kms keys create key-retail-data \
  --location=$REGION \
  --keyring=keyring-retail-g9 \
  --purpose=encryption

# 4. Guardar la ruta de la llave en una variable (la usaremos mucho)
export KMS_KEY="projects/$PROJECT_ID/locations/$REGION/keyRings/keyring-retail-g9/cryptoKeys/key-retail-data"

echo "🔑 Llave creada: $KMS_KEY"
Operation "operations/acet.p2-560566330844-03748ded-b2f1-4903-b67b-0d1802738d0c" finished successfully.
API [cloudkms.googleapis.com] not enabled on project [560566330844]. Would you like to enable and retry (this will take a few minutes)? (y/N)? Y

Enabling service [cloudkms.googleapis.com] on project [560566330844]...
🔑 Llave creada: projects/pc4-si807-g9/locations/us-central1/keyRings/keyring-retail-g9/cryptoKeys/key-retail-data
dlaricacruz@cloudshell:~ (pc4-si807-g9) $

```

Damos permiso a los servicios de Google

```

dlaricacruz@cloudshell:~ (pc4-si807-g9) $ # Obtener emails de servicio
export STORAGE_SA=$(gcloud storage service-agent)
export BQ_SA="bq-$(PROJECT_NUMBER)@bigquery-encryption.iam.gserviceaccount.com"
export DATAPROC_SA="service-$(PROJECT_NUMBER)@dataproc-accounts.iam.gserviceaccount.com"

# Dar permiso de encriptar/desencriptar a Storage
gcloud kms keys add-iam-policy-binding $KMS_KEY \
  --location=$REGION \
  --keyring=keyring-retail-g9 \
  --member="serviceAccount:$STORAGE_SA" \
  --role="roles/cloudkms.cryptoKeyEncrypterDecrypter"

# Dar permiso a BigQuery
gcloud kms keys add-iam-policy-binding $KMS_KEY \
  --location=$REGION \
  --keyring=keyring-retail-g9 \
  --member="serviceAccount:$BQ_SA" \
  --role="roles/cloudkms.cryptoKeyEncrypterDecrypter"

# Dar permiso a Dataproc
gcloud kms keys add-iam-policy-binding $KMS_KEY \
  --location=$REGION \
  --keyring=keyring-retail-g9 \
  --member="serviceAccount:$DATAPROC_SA" \
  --role="roles/cloudkms.cryptoKeyEncrypterDecrypter"

Updated IAM policy for key [key-retail-data].
bindings:
- members:
  - serviceAccount:service-560566330844@gs-project-accounts.iam.gserviceaccount.com
    role: roles/cloudkms.cryptoKeyEncrypterDecrypter
etag: BwZE60gP8J0=
version: 1
Updated IAM policy for key [key-retail-data].
bindings:

```

Encriptamos los servicios

```

dlaricacruz@cloudshell:~ (pc4-si807-g9) $ # 1. Configurar Bucket por defecto con KMS
gsutil kms encryption -k $KMS_KEY gs://$BUCKET_NAME

# 2. Verificar (Toma captura de esto para evidencia)
gsutil kms encryption gs://$BUCKET_NAME
Authorized service account service-560566330844@gs-project-accounts.iam.gserviceaccount.com to use key:
projects/pc4-si807-g9/locations/us-central1/keyRings/keyring-retail-g9/cryptoKeys/key-retail-data
Setting default KMS key for bucket gs://pc4-si807-g9-bucket...
Default encryption key for gs://pc4-si807-g9-bucket:
projects/pc4-si807-g9/locations/us-central1/keyRings/keyring-retail-g9/cryptoKeys/key-retail-data
dlaricacruz@cloudshell:~ (pc4-si807-g9) $

```

VPC

Para crear una red privada donde hacer el procesamiento.

```

dlaricacruz@cloudshell:~ (pc4-si807-g9) $ # 1. Crear la Red VPC (Custom)
gcloud compute networks create vpc-retail-g9 \
  --subnet-mode=custom \
  --bgp-routing-mode=regional

# 2. Crear Subred (con Acceso Privado a Google activado - CRÍTICO para Dataproc)
gcloud compute networks subnets create sub-retail-us \
  --network=vpc-retail-g9 \
  --region=$REGION \
  --range=10.0.0.0/24 \
  --enable-private-ip-google-access
Created [https://www.googleapis.com/compute/v1/projects/pc4-si807-g9/global/networks/vpc-retail-g9].
NAME: vpc-retail-g9
SUBNET MODE: CUSTOM
BGP ROUTING MODE: REGIONAL
IPV4 RANGE:
GATEWAY IPV4:
INTERNAL IPV6 RANGE:

Instances on this network will not be reachable until firewall rules
are created. As an example, you can allow all internal traffic between
instances as well as SSH, RDP, and ICMP by running:

$ gcloud compute firewall-rules create <FIREWALL_NAME> --network vpc-retail-g9 --allow tcp,udp,icmp --source-ranges <IP_RANGE>
$ gcloud compute firewall-rules create <FIREWALL_NAME> --network vpc-retail-g9 --allow tcp:22,tcp:3389,icmp

Created [https://www.googleapis.com/compute/v1/projects/pc4-si807-g9/regions/us-central1/subnetworks/sub-retail-us].
NAME: sub-retail-us
REGION: us-central1
NETWORK: vpc-retail-g9
RANGE: 10.0.0.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL IPV6 PREFIX:
EXTERNAL IPV6 PREFIX:
dlaricacruz@cloudshell:~ (pc4-si807-g9) $

```

Configuración del firewall

```
dlariccoruz@cloudshell:~ (pc4-ai807-g9)$ # 1. Permitir comunicación interna (entre nodos de Dataproc)
gcloud compute firewall-rules create allow-internal-retail \
  --network=vpc-retail-g9 \
  --allow=tcp,udp,icmp \
  --source-ranges=10.0.0.0/24

# 2. Permitir SSH (opcional, buena práctica para debug)
gcloud compute firewall-rules create allow-ssh-retail \
  --network=vpc-retail-g9 \
  --allow=tcp:22 \
  --source-ranges=0.0.0.0/0 \
  --target-tags=ssh-enabled
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/pc4-ai807-g9/global/firewalls/allow-internal-retail].
Creating firewall...done.
NAME: allow-internal-retail
NETWORK: vpc-retail-g9
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp,udp,icmp
DENY:
DISABLED: False
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/pc4-ai807-g9/global/firewalls/allow-ssh-retail].
Creating firewall...done.
NAME: allow-ssh-retail
NETWORK: vpc-retail-g9
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22
DENY:
DISABLED: False
```